Enderton §1.1: The Language of Sentential Logic

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Notes, draft 2

1 Object language and metalanguage

Definition 1.1 (Object language)**.** *Source.* Manin, Part I §1.1; Tarski, Ch. I.

To write: Your definition: what the object language is, what it consists of, and that it is studied rather than used. Include an example and a non-example.

Definition 1.2 (Metalanguage)**.** *Source.* Manin, Part I §1.1; Tarski, Ch. I.

To write: Your definition: the language used to describe the object language, and what lives only here — metavariables such as α, β, the quantifiers of Chapter 1, claims such as "α is a wff". Include an example and a non-example.

Remark (Lean)*.*

To write: One line: which of the two Lean plays, and which an inductive `Formula` type plays.

2 Symbols

Definition 2.1 (Symbols of the sentential language)**.** *Source.* Enderton §1.1.

The symbols are:

Symbol	Name	Kind	Arity
(	left parenthesis	punctuation (logical)	—
)	right parenthesis	punctuation (logical)	—
$\neg$	negation	connective (logical)	1
$\wedge$	conjunction	connective (logical)	2
$\vee$	disjunction	connective (logical)	2
$\rightarrow$	conditional	connective (logical)	2
$\leftrightarrow$	biconditional	connective (logical)	2
$A_1, A_2, \ldots$	sentence symbols	parameter (nonlogical)	0 (our convention)

The *sentential connectives* are $\neg, \wedge, \vee, \rightarrow, \leftrightarrow$. The *logical symbols* are the parentheses and the connectives. The *parameters* (nonlogical symbols) are the sentence symbols. There are infinitely many sentence symbols.

Quantified form.

> **To write:** Test 7 (structure): write the signature of a sentential language, its components and their types.

Convention 2.2 (Ours, not Enderton's)**.** We record sentence symbols as having arity 0. Parentheses have no arity: they are punctuation, not operations.

Assumption 2.3 (Distinctness)**.** *Source.* Enderton §1.1.
The symbols listed are pairwise distinct; for example, $A_3 \neq \leftrightarrow$.

Assumption 2.4 (No symbol is a sequence of others)**.** *Source.* Enderton §1.1.
No symbol is a finite sequence of two or more symbols; for example, $A_3 \neq \langle \neg, A_4, (\, \rangle$.

Example 2.5 (Why symbols must be atomic: `A1` inside `A11`)**.** Suppose sentence symbols were written as strings of characters, so that A_1 is the string `A1` and A_{11} is the string `A11`. Then `A1` occurs inside `A11`, and replacing every occurrence of `A1` by `B` would turn the expression `(A1 -> A11)` into `(B -> B1)`: the replacement corrupts A_{11}, which never contained A_1 as a symbol.

Under Assumption 2.4, A_{11} is a single atomic symbol, not a sequence with A_1 as a part. An occurrence of A_1 in an expression is therefore an entry of the sequence equal to A_1, and replacing occurrences of A_1 cannot touch A_{11}.

Remark. This matters for *substitution* (replacing every occurrence of a sentence symbol in a wff by a wff), which comes later. It is unrelated to *concatenation*, e.g. $(\alpha \to \beta) = \mathcal{E}_{\to}(\alpha, \beta)$, which places whole expressions into a fixed pattern and replaces nothing.

Proposition 2.6 (Unique decomposition of expressions)**.** *Source. Enderton §1.1.*
A finite sequence of symbols can be decomposed in only one way.

Quantified form.

If $\langle a_1, \ldots, a_m \rangle = \langle b_1, \ldots, b_n \rangle$, then $m = n$ and $a_i = b_i$ for each i.

> **To write:** Missing hypothesis from the review: add the condition on the a_i and b_j without which this is false.

Study question. Why is this not immediate from equality of tuples? Look at how Enderton defines n-tuples in Chapter 0.

3 Expressions

Definition 3.1 (Expression)**.** *Source.* Enderton §1.1.
An *expression* is a finite sequence of symbols. We write Expr for the set of expressions.

Convention 3.2 (Notation for expressions)**.** We *denote* an expression by writing the names of its symbols side by side. For example, $(\neg A_1)$ denotes $\langle (\,, \neg, A_1, \,) \rangle$. If α and β denote expressions, $(\alpha \to \beta)$ denotes their concatenation with the symbols shown.

Example 3.3. Let $\alpha = (\neg A_1)$ and $\beta = A_2$. Then $(\alpha \to \beta) = ((\neg A_1) \to A_2)$.

Example 3.4 (Non-example of a formula)**.** The expression $((\to A_3$ is an expression but is nonsense; a logic must exclude such expressions.

Remark (Translation). An English sentence such as "Roses are red" is presumably true or false. Its translation is a sentence symbol, or for a compound sentence an expression; truth values enter only in §1.2.

4 Well-formed formulas, informally

Definition 4.1 (Formula-building operations)**.** *Source.* Enderton §1.1.

For expressions α, β:

$$\mathcal{E}_\neg(\alpha) = (\neg\alpha), \qquad \mathcal{E}_\wedge(\alpha,\beta) = (\alpha \wedge \beta), \qquad \mathcal{E}_\vee(\alpha,\beta) = (\alpha \vee \beta),$$
$$\mathcal{E}_\to(\alpha,\beta) = (\alpha \to \beta), \qquad \mathcal{E}_\leftrightarrow(\alpha,\beta) = (\alpha \leftrightarrow \beta).$$

Definition 4.2 (Well-formed formula, informal)**.** *Source.* Enderton §1.1.

(a) Every sentence symbol is a wff.

(b) If α and β are wffs, then so are

> **To write:** the five expressions produced by the formula-building operations.

(c) No expression is a wff unless it is compelled to be one by (a) and (b).

Remark. Clause (c) is not yet rigorous. Section 5 makes it so. A wff can also be pictured as a tree, e.g. A_1, A_2 under $\wedge$ giving $(A_1 \wedge A_2)$; Enderton gives the rules for trees later (§1.3).

5 Making clause (c) rigorous

Definition 5.1 (Construction sequence)**.** *Source.* Enderton §1.1.

A *construction sequence* is a finite sequence $\langle \varepsilon_1, \ldots, \varepsilon_n \rangle$ of expressions such that for each $i \le n$ at least one of the following holds:

(i) ε_i is a sentence symbol;

(ii) $\varepsilon_i = \mathcal{E}_\neg(\varepsilon_j)$ for some $j < i$;

(iii) $\varepsilon_i = \mathcal{E}_\square(\varepsilon_j, \varepsilon_k)$ for some $j < i$, $k < i$, where $\square$ is one of $\wedge, \vee, \to, \leftrightarrow$.

Quantified form.

> **To write:** Test 6 (predicate): "$\langle \varepsilon_1, \ldots, \varepsilon_n \rangle$ is a construction sequence" $\iff$ ...

Remark (Duplicates and unused entries). A construction sequence may contain repeated entries and entries that are not used to build the last one, so a wff has many construction sequences.

> **To write:** Two examples that satisfy (i)–(iii): one with a duplicate, one with an unused entry. Check ε_1 first.

Remark (Our generalization). One could allow connectives of any arity n, with clause (iii) reading $\varepsilon_i = \mathcal{E}_\square(\varepsilon_{j_1}, \ldots, \varepsilon_{j_n})$ with every $j_r < i$. Whether an n-ary connective can be expressed by binary ones is a semantic question, deferred to §1.5.

Definition 5.2 (Wff, bottom-up)**.** *Source.* Enderton §1.1.

An expression α is a wff if

To write: Test 4 (quantifier): which quantifier on construction sequences? Not "the".

Definition 5.3 (Closed set)**.** *Source.* Enderton §1.1, p. 18.

To write: Test 6 (predicate): "S is closed under the formula-building operations" $\iff$ ... (all five operations, negation included).

Remark (Looking ahead to §1.4). Enderton's §1.1 characterizes wffs bottom-up only (Definition 5.2). In §1.4 he gives a second definition "from the top down", as the intersection of all *inductive* sets, and proves the two agree; construction sequences are the "bottom up" one. Those nodes belong to the §1.4 notes.

6 Induction

Theorem 6.1 (Induction principle)**.** *Source. Enderton §1.1, p. 18.*

If S is a set of wffs containing all the sentence symbols and closed under the formula-building operations, then S is the set of all wffs.

Quantified form.

To write: Test 1: the quantifier over S, with its domain named.

Remark (Proof structure). The conclusion is a set equality, $S = \mathsf{WFF}$. One inclusion is free: $S \subseteq \mathsf{WFF}$ is a hypothesis. So the work is $\mathsf{WFF} \subseteq S$: take an arbitrary wff α and show $\alpha \in S$.

The only handle on an arbitrary wff is the bottom-up definition: some construction sequence ends with α. Walk along that sequence by strong induction, showing every entry lies in S; the last entry is α.

The induction has the shape of the definition of a construction sequence. The first entry has nothing before it, so it can only be a sentence symbol: sentence symbols must appear before anything is built from them. Every later entry is either a sentence symbol, in S by hypothesis, or is built from earlier entries by a formula-building operation, in S by the inductive hypothesis and closure.

Proof. By hypothesis $S \subseteq \mathsf{WFF}$, so it suffices to show $\mathsf{WFF} \subseteq S$. Let α be an arbitrary wff; we show $\alpha \in S$.

Since α is a wff, there is a construction sequence $\langle \beta_1, \ldots, \beta_n \rangle$ with $\beta_n = \alpha$ (Definition 5.2). We prove by strong induction on $k \in \{1, \ldots, n\}$ that $\beta_k \in S$.

Base case, $k = 1$. There are no entries before β_1, so clauses (ii) and (iii) of Definition 5.1 cannot hold; hence β_1 is a sentence symbol. Since S contains every sentence symbol, $\beta_1 \in S$.

Inductive case, $k > 1$. Assume $\beta_i \in S$ for all $i < k$. By Definition 5.1, at least one of the following holds for β_k:

(i) β_k is a sentence symbol. Then $\beta_k \in S$ by hypothesis.

(ii) $\beta_k = \mathcal{E}_\neg(\beta_j)$ for some $j < k$. By the inductive hypothesis $\beta_j \in S$, and S is closed under $\mathcal{E}_\neg$, so $\beta_k \in S$.

(iii) $\beta_k = \mathcal{E}_\square(\beta_j, \beta_l)$ for some $j < k$, $l < k$ and some binary connective $\square \in \{\wedge, \vee, \to, \leftrightarrow\}$. By the inductive hypothesis $\beta_j, \beta_l \in S$, and S is closed under $\mathcal{E}_\square$, so $\beta_k \in S$.

In every case $\beta_k \in S$. By strong induction, $\beta_k \in S$ for all $k \leq n$; in particular $\alpha = \beta_n \in S$.

Since α was arbitrary, $\mathsf{WFF} \subseteq S$, and with $S \subseteq \mathsf{WFF}$ we conclude $S = \mathsf{WFF}$. □

Remark (Idea). A wff is the last line of a construction sequence, and S absorbs every line in turn.

Definition 6.2 (Parenthesis counts)**.** For an expression ε, let $l(\varepsilon)$ be the number of occurrences of the symbol "(" in ε, and $r(\varepsilon)$ the number of occurrences of ")". In particular $l(() = 1$, $r()) = 1$, and $l(s) = r(s) = 0$ for every other symbol s.

Remark. l and r are functions $\mathsf{Expr} \to \mathbb{N}$, not counters: every expression already has a value, and proofs compute those values rather than incrementing anything.

Lemma 6.3 (Counts add under concatenation)**.** *For all expressions ε, δ: $l(\varepsilon\delta) = l(\varepsilon) + l(\delta)$ and $r(\varepsilon\delta) = r(\varepsilon) + r(\delta)$.*

Proof.

To write: Your proof (short: occurrences in $\varepsilon\delta$ are the occurrences in ε together with those in δ).

□

Remark (Structure). With concatenation, Expr is a monoid whose identity is the empty sequence; it is the free monoid on the symbols. The lemma says l and r are monoid homomorphisms $(\mathsf{Expr}, \cdot, \langle\rangle) \to (\mathbb{N}, +, 0)$, so each is determined by its values on single symbols.

Theorem 6.4 (Unbalanced parentheses)**.** *Source. Enderton §1.1, p. 18 (stated there as an Example).*

Any expression with more left parentheses than right parentheses is not a wff.

Quantified form.

To write: Test 5: make the hidden ∀ explicit.

Remark (Proof structure). Prove something stronger: *every* wff is balanced, $l(\alpha) = r(\alpha)$. The formation rules force this, because each one wraps balanced inputs in exactly one "(" and one ")". Make that precise by strong induction along a construction sequence of an arbitrary wff. The theorem is then the contrapositive.

Proof. Let α be a wff. By Definition 5.2 there is a construction sequence $\langle \beta_1, \ldots, \beta_n \rangle$ with $\beta_n = \alpha$. We show by strong induction on $k \in \{1, \ldots, n\}$ that $l(\beta_k) = r(\beta_k)$.

Base case, $k = 1$. As in the proof of Theorem 6.1, β_1 is a sentence symbol, so $l(\beta_1) = 0 = r(\beta_1)$.

Inductive case, $k > 1$. Assume $l(\beta_i) = r(\beta_i)$ for all $i < k$. By Definition 5.1, one of the following holds.

(i) β_k is a sentence symbol. Then $l(\beta_k) = 0 = r(\beta_k)$.

(ii) $\beta_k = \mathcal{E}_\neg(\beta_j) = (\neg\beta_j)$ for some $j < k$. By Lemma 6.3,

$$l(\beta_k) = 1 + 0 + l(\beta_j) + 0, \qquad r(\beta_k) = 0 + 0 + r(\beta_j) + 1,$$

and $l(\beta_j) = r(\beta_j)$ by the inductive hypothesis, so $l(\beta_k) = r(\beta_k)$.

(iii) $\beta_k = \mathcal{E}_\square(\beta_j, \beta_m) = (\beta_j \,\square\, \beta_m)$ for some $j, m < k$ and binary connective $\square$. By Lemma 6.3,

$$l(\beta_k) = 1 + l(\beta_j) + 0 + l(\beta_m) + 0, \qquad r(\beta_k) = 0 + r(\beta_j) + 0 + r(\beta_m) + 1,$$

and $l(\beta_j) = r(\beta_j)$, $l(\beta_m) = r(\beta_m)$ by the inductive hypothesis, so $l(\beta_k) = r(\beta_k)$.

By strong induction $l(\beta_k) = r(\beta_k)$ for all $k \leq n$; in particular $l(\alpha) = r(\alpha)$.

So every wff has $l = r$. An expression with $l > r$ therefore is not a wff. □

Remark (Alternative proof via the induction principle). Let $S = \{\alpha \in \mathsf{WFF} : l(\alpha) = r(\alpha)\}$. Then $S \subseteq \mathsf{WFF}$ by definition, and every sentence symbol is in S. If $\alpha, \beta \in S$, the computations in cases (ii) and (iii) above show $(\neg\alpha) \in S$ and $(\alpha \,\square\, \beta) \in S$, so S is closed under the five operations. By Theorem 6.1, $S = \mathsf{WFF}$. The construction sequence is not mentioned: the strong induction along it was done once, inside the proof of the induction principle.

Remark. The proof shows more than the statement: no wff has more *right* parentheses than left either. Any mismatch rules an expression out.

Study question. Is the converse true? If not, give a counterexample.